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## Identification of Functional Groups of Sustainable Bio-oil Substrate and Its Potential for Specialty chemicals Source

Nasri N. Shawal<sup>1,a</sup>, Ahmed M. Murtala<sup>1,3,b</sup>, Abdul K. Adilah<sup>1,c</sup>,  
and Hamza D. Usman<sup>2,d</sup>

<sup>1</sup>Sustainable Waste To Wealth of Gas Technology Centre, Faculty of Petroleum and Renewable Energy Engineering, University Teknologi Malaysia, 81310 UTM Johor Bahru, Johor, Malaysia

<sup>2</sup>Faculty of Chemical Engineering, Universiti Teknologi Malaysia, 81310 UTM Johor Bahru, Johor, Malaysia

<sup>3</sup>Department of Chemical Engineering, Faculty of Engineering, University of Maiduguri, P.M.B. 1069 Maiduguri, Borno, Nigeria

<sup>a</sup>noorshaw@petroleum.utm.my, <sup>b</sup>muriahmed@yahoo.com, <sup>c</sup>adilahadulkadir@gmail.com, <sup>d</sup>usmandhamza@yahoo.com

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**Abstract.** This paper describes the basic analytical technique used in characterizing bio-oil into various components in terms of function groups as well as assessing its thermal behaviour. Pyrolysis, being one of the most widely used techniques in converting biomass was employed to get bio-oil from empty palm fruit bunches (EPFB). The bio-oil obtained was characterized using Fourier Transform Infrared (FTIR) while the thermal behaviour were assessed using Thermogravimetric analysis (TGA) and Differential Scanning Calometry (DSC). The results showed a great range of functional groups of phenol, alcohols, ketones, aldehydes, carboxylic acids, alkanes, alkenes and aromatic in the FTIR spectrum. The onset thermal temperature of thermal decomposition from TGA result is 32°C corresponding to about 13.4281 mg and the amount of residue was found to be 3.9969mg which shows that the bio-oil losses weight very fast; a characteristic of lighter product which strongly support combustion. The stages on DTG curve, in the degradation, which occurred at temperature range between 70-90°C, correspond to the removal of light hydrocarbons (such as aldehydes, alcohols and carboxylic acids) and moisture content in the bio-oil. The glass transition temperature, T<sub>g</sub>, was found to be 53.63°C indicating that the bio-oil tested is less rigid and hence requires less force delivery. The characterization results indicate that the bio-oil obtained from oil palm EFB can be a potentially valuable source of fuels and chemicals feedstocks. Utilization of bio-oil for fuels and chemicals manufacture can significantly reduce or eliminate the harmful effects of fossil based products on the environment.

### Introduction

The need for alternative sources of energy to replace fossil based primary feedstock is attracting more attention and playing an important role in the world energy economy in recent times. Renewable resources such as biomass is a promising option that could significantly leads to important environmental benefits, particularly to the reduction of atmospheric CO<sub>2</sub> concentrations and, thereby of the greenhouse effect [1].

Biomass mainly refers to any organic matter derived from agricultural or forestry sector that is available on a recurring basis (sustainable). It comprises of C, H, O and N; a composition similar to fossil feedstock which has C and H [2,3,4,5]. As such similar products can be obtained from biomass with that of petroleum counterparts [3,4,6]. Carbohydrates (which are the currently limited non-food uses for starch), cellulose and hemicelluloses are the largest renewable sources of carbon [5,7].

Application of new and advanced technologies in fuel and chemical processing provide new opportunities for the conversion of these enormous natural resources (biomass). Pyrolysis, being one of the most widely used techniques in converting biomass is widely employed to get bio-oil, char and gas depending on the pyrolysis conditions [8,9]. Bio-oil is a polar and high density oxygenated liquid that can be used as a fuel and for the production of chemicals [10,11]. However, a complete analysis of bio-oil composition required a set of analytical techniques.

Component characterization of bio-oil compounds into various functional groups as well as the assessment of its thermal behaviour are very useful and necessary because various components of the bio-oil can be treated as a mixture of few groups instead of hundreds of compounds [9,12,13,14]. Some of the promising techniques used include Fourier transform infrared (FTIR), Thermogravimetric analysis (TGA), Differential scanning calorimetry (DSC) [9,13].

This article is aimed at describing the basic analytical technique used in determining the various components (functional groups) of bio-oil obtained from the pyrolysis of EFB with the view of assessing its thermal behaviour towards provision of a sustainable and valuable compound.

## Experimental Methods

The EPFB used for the experiments was received in the form of whole bunches and in a fairly dry condition with less than 10 wt %. The bunches were then chopped into smaller sizes, and subsequently, a Fritsch grinder with a screen size of 500  $\mu\text{m}$  was used to reduce the size of the feedstock to less than 500  $\mu\text{m}$  with the particle sizes of interest for this study being 250  $\mu\text{m}$ , as the feedstock of this size range can easily be fed into the reactor.

### 2.1 Bio-oil Sample Preparation

Pyrolysis of EPFB was carried out using a packed-bed reactor which consists of essentially, a quartz tube (height 390 mm), a cooling system for the separation of water and condensable organic vapors (tar) and a gas cleaning/drying system followed by gas-measurement devices. For every trial, 5 g of the sample was placed into the reactor prior to heating. Fig. 1 shows the schematic diagram of the EPFB pyrolysis test rig facility. Inert purging gas (pure  $\text{N}_2$ ) was provided from the bottom of the reactor at  $4.5\text{L/min} \pm 10\text{ mL/min}$  to maintain the pyrolysis atmosphere. The furnace used was heated to  $525^\circ\text{C} \pm 25^\circ\text{C}$  with the heating rate  $10^\circ\text{C/min}$ . The pyrolysis process was timed for 3hrs and the volatile products were cooled and collected for further analysis.

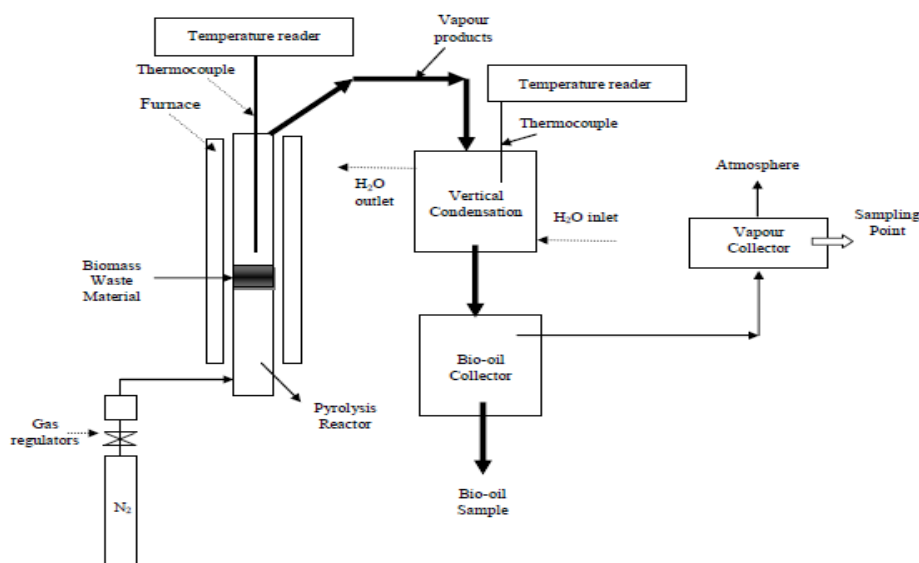


Fig. 1: Schematic diagram of EPFB pyrolysis process in a batch reactor.

## 2.2 Bio-Oil Characterization

2.2.1 Thermogravimetric analysis (TGA): is the most widely used thermal method. It is based on the measurement of mass loss of material as a function of temperature. The thermal behaviour of the bio-oil was investigated using Mettler Toledo TGA/DSC1 thermogravimetric analyzer with a heating rate of 10°C/min at a temperature range of 25 – 400°C. The TGA run was performed under a nitrogen purge. Fig. 2 shows the TGA and DTG curves.

2.2.2 Differential scanning calorimetry (DSC): is a thermal analysis technique which measures the temperature and heat flow associated with transitions in materials as a function of temperature and time. Such measurements provide quantitative and qualitative information about physical and chemical changes that include endothermic/exothermic processes or changes in heat capacity. This analysis was conducted using Mettler Toledo TGA/DSC1 instrument with a heating rate of 10°C/min. Fig. 3 shows the DSC curve.

2.2.3 FTIR analysis: FTIR spectroscopic analysis was performed using a Perkin Elmer Spectrum One series model instrument. Analysis of FTIR spectra at wave numbers (ranging between 350-4000cm<sup>-1</sup>) of the bio-oil provides a quick and simple qualitative technique that uses the standard IR spectra to identify the functional group(s) of the components of the bio-oil. The results from the analysis of Fig. 4 are summarized in Table 1 below.

## Results and Discussion

### 3.1 TGA

Thermal stability is an important factor that can be used to select material for certain end-use, predict product performance and improve product quality. From the TGA curve, the results indicate that the bio-oil decomposes in a single step between 32°C and 140°C. The onset thermal temperature of thermal decomposition is therefore 32°C corresponding to about 13.4281 mg and the final decomposition temperature is 140°C with the amount of residue being 3.9969mg. This shows that the bio-oil losses weight very fast; a characteristic of lighter product. Hence it can strongly support combustion [15].

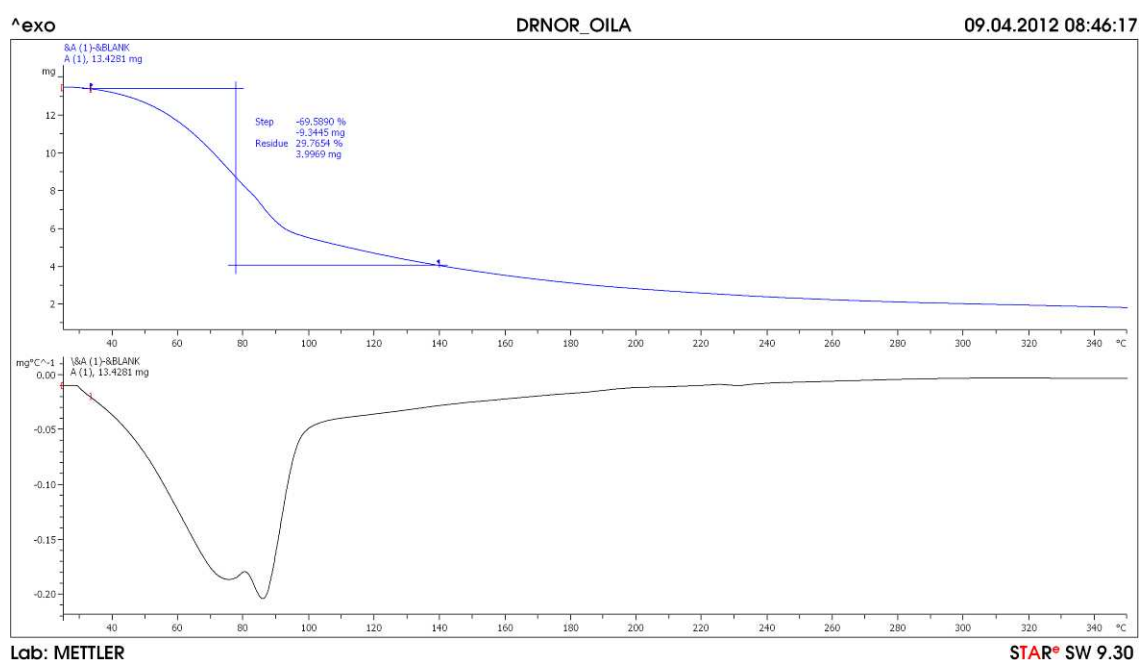


Fig. 2: TGA and DTG curves of bio-oil at a heating rate of 10°C/min.

The stages on DTG curve, in the degradation, which occurred at temperature range between 70-90°C, correspond to the removal of light hydrocarbons (such as aldehydes, alcohols and carboxylic acids) and moisture content in the bio-oil. This also indicated at temperature lower than 100°C, the volatilization absorbed a great deal of heat [16].

### 3.2 DSC

The information these instruments generate is used to understand amorphous and crystalline behavior, polymorph and eutectic transitions, curing and degree of cure, and many other material properties used to design, manufacture and test products. Glass transition temperature ( $T_g$ ) is one of the fundamental properties which provide an insight into the molecular structure which point out to the overall clinical performance of the material. A higher  $T_g$  suggests a material to be in an earlier region of the rubbery plateau at room temperature which indicates a more rigid material. The greater the rigidity of the material the more will be the force delivery of the material.

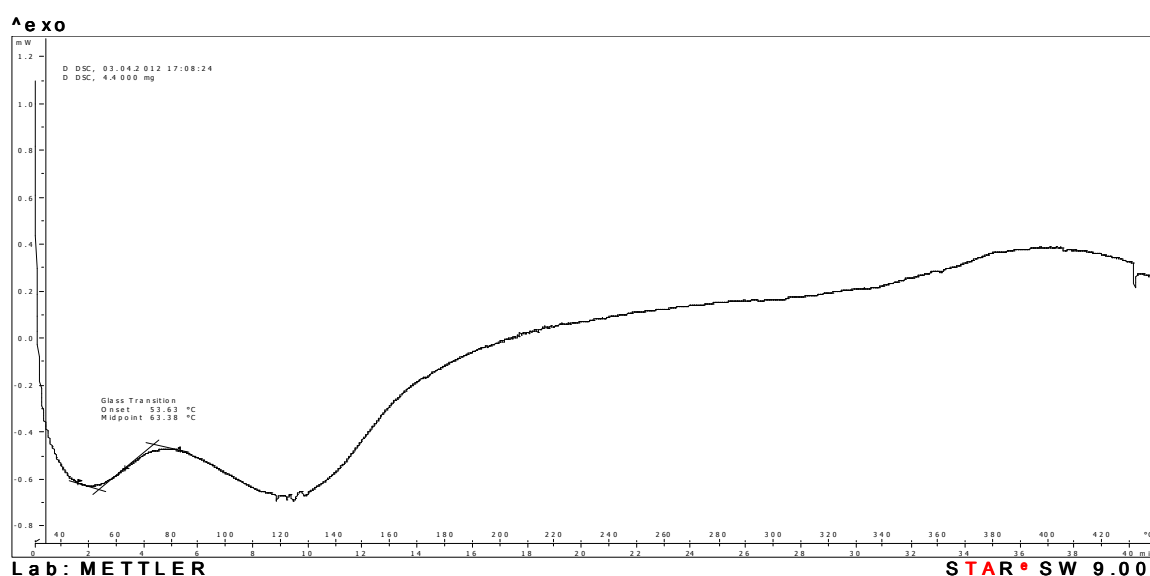


Fig. 3: DSC thermogram of bio-oil under heating rate of 10°C/min.

From DSC curve, the results indicate the presence of a typical endothermic peak in the range of 60-90°C and the corresponding onset  $T_g$  is 53.63°C with the midpoint  $T_g$  as 63.38°C. Therefore the bio-oil tested is less rigid and hence require less force delivery. Storage at temperatures just below  $T_g$  permit development of order as the sample moves towards equilibrium. The molecular motion associated with the glass transition is time dependent. Therefore,  $T_g$  increases when heating rate increases.

### 3.3 FTIR

The FTIR spectra, representing functional group of bio-oil obtained from the pyrolysis of EPFB is shown in Fig. 4. The broad peak of the O–H stretching vibrations between 3600 and 3200  $\text{cm}^{-1}$  indicates the presence of alcohols or more specific phenols (at 3417.87 $\text{cm}^{-1}$ ) in the bio-oil [9]. The C=O stretching vibrations between 1725-1705  $\text{cm}^{-1}$  indicates the presence of carboxylic acids and ketones. The C–O stretching vibration between 3000-1000  $\text{cm}^{-1}$  indicates the presence of alcohol and carboxylic acids. The presence of alkane groups in the bio-oil derived from the kernel was indicated by the C–H stretching vibrations between 3000 and 2850  $\text{cm}^{-1}$ . In addition, the location of bending vibration of C–H groups at 1385.50  $\text{cm}^{-1}$  provides another evidence for the presence of methyl groups in a given compound. Alkenes presence was indicated by C=C stretching vibrations between 1680-1600  $\text{cm}^{-1}$ .

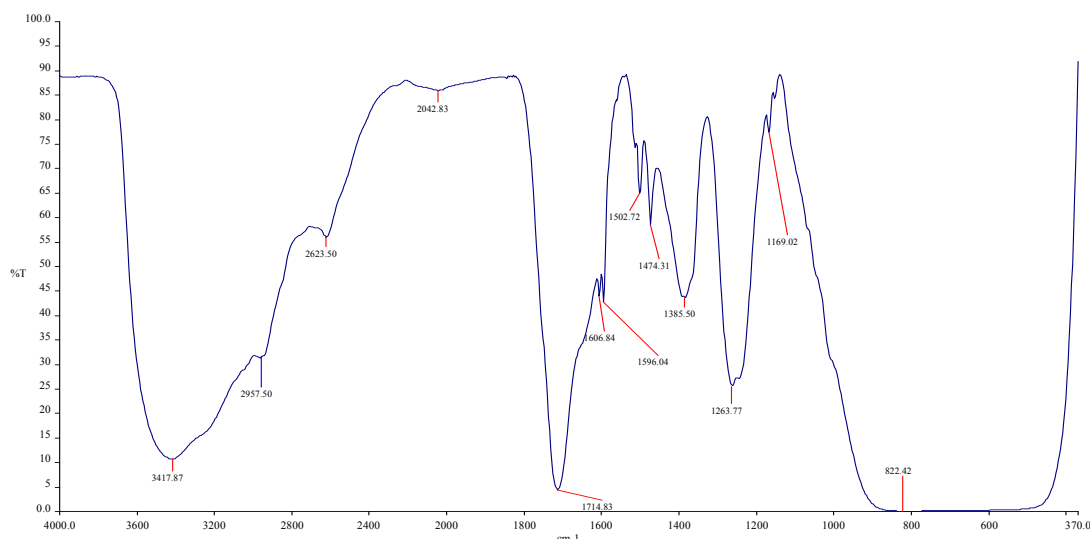


Fig. 4: FTIR spectra of the bio-oil.

Finally aromatics presence was indicated by C=C stretching vibrations between 1600-1470  $\text{cm}^{-1}$  and 690-690  $\text{cm}^{-1}$  [15,17,18]. The above analysis can be summarized in Table 1.0 below.

Table 1: FTIR functional group composition analysis of the bio-oil obtained from EFB [15,17,18].

| Wave number range ( $\text{cm}^{-1}$ ) | Peak wave number( $\text{cm}^{-1}$ ) | Group           | Compound                             |
|----------------------------------------|--------------------------------------|-----------------|--------------------------------------|
| 3600-3200                              | 3417.87                              | O-H stretching  | Alcohols , Phenols                   |
| 3000-2850                              | 2957.50                              | C-H stretching  | Alkanes                              |
| 3000-1000                              | 1263.77                              | C-O stretching  | Alcohols , carboxylic acids          |
| 1725-1705                              | 1714.83                              | C=O stretching  | Carboxylic acids, Aldehydes, Ketones |
| 1680-1600                              | 1606.84                              | C=C stretching  | Alkenes                              |
| 1600-1470                              | 1596.04,1502.72,1474.31              | C=C stretching  | Aromatics                            |
| 1450-1375                              | 1385.50                              | C-H bending     | Alkanes                              |
| 960-600                                | 822.42                               | C-H deformation | Aromatics                            |

The above results in Table 1 can be compared with the results of other studies using both EFB and other raw materials as presented in Table 2 below:

Table 2: Comparison of bio-oil functional groups from different raw materials.

| Wave Number range | Group           | compound                             | Raw material       |          |          |                 |
|-------------------|-----------------|--------------------------------------|--------------------|----------|----------|-----------------|
|                   |                 |                                      | EFB [present work] | EFB [10] | EFB [12] | Cottonseed [19] |
| 3600-3200         | O-H stretching  | Alcohol, Phenol                      | X                  | -        | X        | X               |
| 3000-2850         | C-H stretching  | Alkanes                              | X                  | X        | X        | X               |
| 3000-1000         | C-O stretching  | Alcohol, Carboxylic acids            | X                  | X        | X        | -               |
| 1725-1705         | C=O stretching  | Carboxylic acids, Aldehydes, Ketones | X                  | X        | X        | X               |
| 1680-1600         | C=C stretching  | Alkenes                              | X                  | X        | X        | X               |
| 1600-1470         | C=C stretching  | Aromatics                            | X                  | -        | X        | -               |
| 1450-1375         | C-H bending     | Alkanes                              | X                  | X        | X        | -               |
| 960-600           | C-H deformation | Aromatics                            | X                  | X        | X        | X               |

## Conclusion

Component characterization of bio-oil compounds into various functional groups as well as assessment of its thermal behaviour are very useful and necessary because various components of the bio-oil can be treated as a mixture of few groups instead of hundred of compounds. These kinds of description can be useful in the modeling bio-oils evaporation and combustion as well as in the development of bio-oil up-grading strategies. From the thermogravimetric analysis, it was found that the bio-oil lost weight very fast; a characteristic of lighter product which strongly support combustion. Thus the bio-oil obtained from oil palm EPFB can be a potentially valuable source of fuels and chemicals and its utilization for fuels and chemicals manufacture can significantly reduce or eliminate the harmful effects of fossil based products on the environment.

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